

Technical Publication

Direction 5114494-100

Revision 01

RIP Motorola Board Replacement

SERVICE DOCUMENTATION

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LANGUAGE

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REVISION HISTORY

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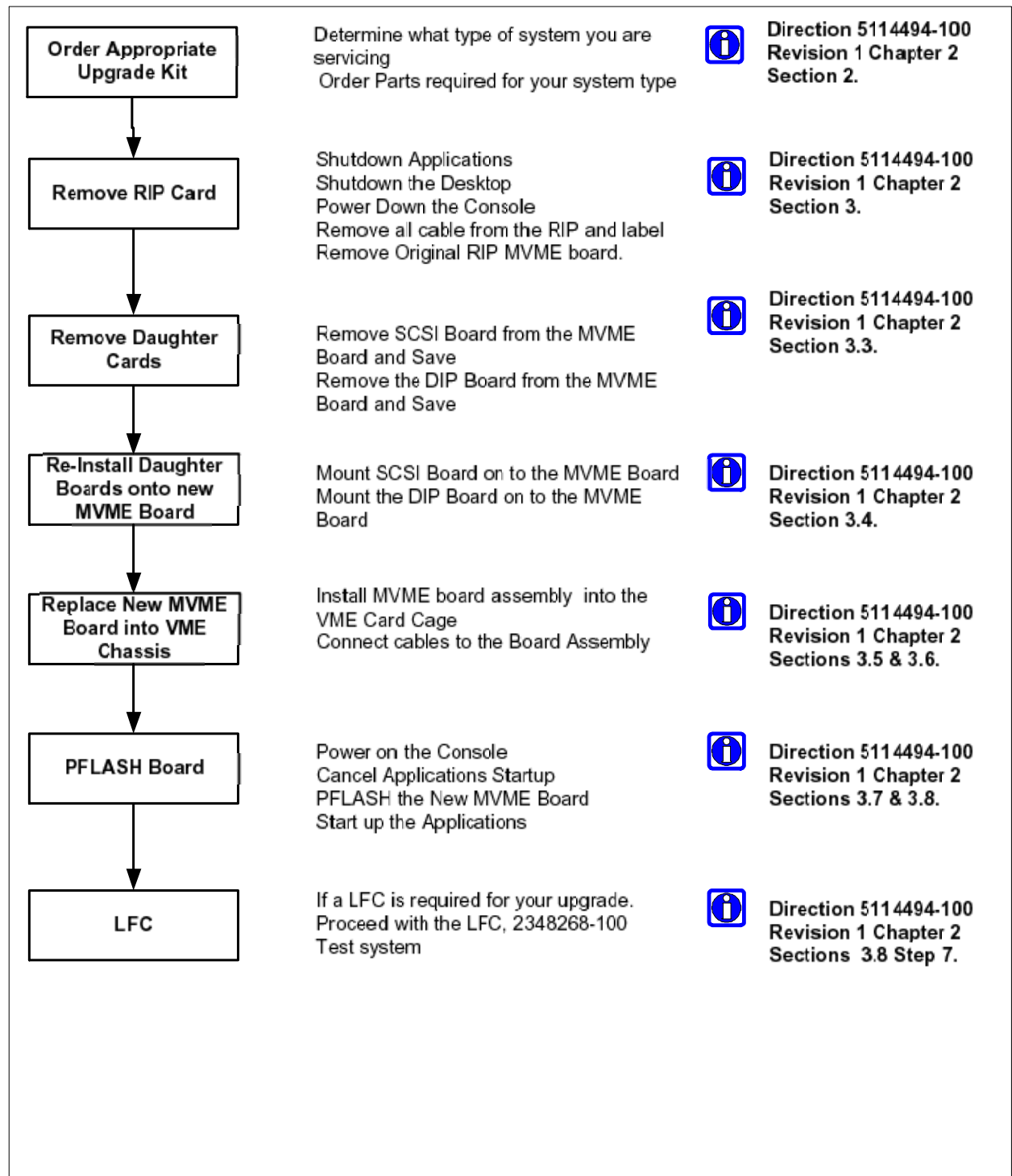
Revision	Date	Reason for change
01	09/02/04	Initial Release

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CHAPTER 1 – STEERING GUIDE

The following Steering Guide is to be used for the MVME Replacement procedure for ALL CT scanners using RIP Board P/N 2339284 and its successor 2370825. (This procedure applies only to CT-only scanners – it does NOT apply to Discovery LS.)



CHAPTER 2 - RIP MOTOROLA BD REPLACEMENT

SECTION 1 - PERSONNEL REQUIREMENTS

Required Persons	Procedure	Finalization
1	45 min	

NOTE: If SW LFC procedure needs to be completed, then plan for one complete day (8hr)

SECTION 2 - PRELIMINARY REQUIREMENTS

2.1 PARTS ORDERING

- Find the system type you are servicing in the table below.

Using the top row, select the system and version of software currently loaded on your CT system. Given this selection, order the “X’ed” parts listed in that column. These parts support the replacement of your MVME2400 SBC (GEMS P/N 2339284) replacement with MVME2400 SBC (GEMS P/N 2370825).

▼ → Find System	IRIX-Based Systems						Linux-Based Systems		
	HS QX/I	LS 3X	LS 4X	LS 4X	LS 3X	LS 4X	HS QX/I (Linux)	LS 3X (Linux)	LS 4X (Linux)
	4 Slice S/W Version 101I.4_HSQXIM4	4 or 8 Slice S/W Version 308I.2_H3.1M5	16 Slice S/W Version 405I.2_H4.0M5	16 Slice S/W Version 405I.3_H4.0M5.1	4 or 8 Slice S/W Version 305.4_H3.1M4	16 Slice S/W Version 401.8_H4.0M4	4 Slice S/W Version 102L.1_HSQXILINUXM3	4 or 8 Slice S/W Version 308L.5_H3.3M4	16 Slice S/W Version 408L.5_H4.2M4
Needed Parts									
MVME 2400 Kit (SBC + Install Doc) 5114901	X	X	X	X	X	X	X	X	X
HighSpeed QXI (IRIX) S/W 2401246 (FMI 25902) 101I.2_HSQXIM5	X								
LightSpeed 3X (IRIX) S/W 2395177 308I.2_H3.1M5					X				
LightSpeed 4X (IRIX) S/W 5116050 405I.3_H4.0M5.1			X			X			
HighSpeed QXI (Linux) S/W 2401245 103L.4_HSQXIM4							X		
LightSpeed 4X (Linux) S/W 5111370 (FMI 25338) 405L.8_H4.2M4									X
HighSpeed QX/I (IRIX) S/W LFC Procedure 2340896-100	X								
LightSpeed 3X & 4X (IRIX) S/W LFC Procedure 2348269-100			X		X	X			
HighSpeed QX/I (Linux) S/W LFC Procedure 2379286-100							X		
LightSpeed 3X & 4X (Linux) S/W LFC Procedure 2362590-100									X

2. To determine if the software release is loaded on the system, do the following:
 - a. Open a Unix Shell.
 - b. Type: **showprods | grep *LightSpeed***
3. If you needed to order the SW LFC procedure, then you **must** proceed with the SW LFC.

2.2 - TOOLS AND TEST EQUIPMENT

Screw Driver

2.3 - REQUIRED CONDITIONS

ESD-Safe Environment.

SECTION 3 - PROCEDURE

NOTE: There are Sub Parts attached or connected to the RIP board (e.g., DIP daughter-card, SCSI daughter). These parts must be removed/detached from the SBC and stored properly (I.E. ESD bags). They will be later reconnected to the replacement RIP Board.

NOTE: After the RIP Board has been replaced in the Operator console, the Pflash procedure needs to be performed.

3.1 - POWER DOWN (OFF) THE CONSOLE

1. Power off the Operator Console following one of the two methods:
 - a. If Applications are up:
 - i. Select the Shutdown Icon, 
 - ii. Select [**SHUTDOWN**]
 - iii. Select [**OK**]
 - b. When the console is shutdown, a message on the Operator Console monitor is displayed:

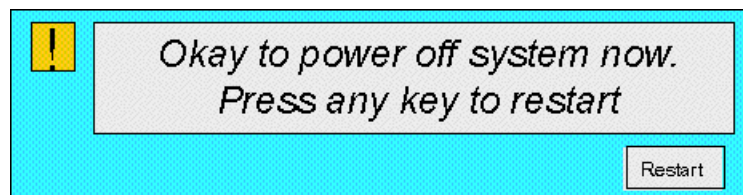


Figure 1 – Message Operator Console Monitor

- c. If Applications are down:

Open a Unix Shell from the Tool chest and type the following:

 - i. **su – <enter>**
 - ii. **password: #bigguy <enter>**
 - iii. **shutdown**

The Operator Console monitor will display a 'Power down' message.

2. Power **OFF** the console at the front switch.

3.2 - REMOVE ORIGINAL RIP MVME MOTOROLA BOARD

1. Remove the front Operator Console cover. For detailed instructions on how to do this, refer to:
 - Direction 2243314 **LS2.x** System Service Manual, Book 3, Chapter 5, Section 3.2.1
 - Direction 2399875-800 **LS 3.x** Service Methods -> *Parts Replacement* -> *Console (GOC1 or GOC2)* -> *SRU Replacements - Global Consoles*-> [Access to the SRU's Interior](#)
 - Direction 5112256-800 **LS4.x** Service Methods -> *Parts Replacement* -> *Console (GOC1 or GOC2)* -> *SRU Replacements - Global Consoles* -> RIP Board Replacement
2. Remove all cable connections to the MVME RIP / Motorola Board. For detailed instructions on how, refer to:
 - Direction 2243314 LS2.x System Service Manual, Book 3, Chapter 5, Section 3.2.1
 - Direction 2399875-800 **LS 3.x** Service Methods -> *Parts Replacement* -> *Console (GOC1 or GOC2)* -> *HTSRU Replacements - Global Consoles* -> [RIP Board Replacement](#)
 - Direction 5112256-800 **LS4.x** Service Methods -> *Parts Replacement* -> *Console (GOC1 or GOC2)* -> *SRU Replacements - Global Consoles* -> RIP Board Replacement
3. Verify each cable label is present and clearly marked. If necessary, add a label for clarity (to make reassembly easier) .
 - RJ45 DEBUG Cable
 - RJ45 10/100 BaseT Cable
 - PCI Mezzanine Card SCSI Daughter Board Cable A
 - DIP Board Daughter Board Fiber optic RX Cable
 - DIP Board Daughter Board DIP Scan Abort Cable
4. Place the original Motorola Board on a static mat.

3.3 - REMOVE DAUGHTER CARDS

3.3.1 - REMOVING SCSI BOARD FROM MOTOROLA BOARD

NOTE: Do not follow steps in the Illustration. This Illustration shows the proper card placement and screw locations. DO use the following steps:

- Direction 2243314 **LS2.x** System Service Manual, Book 3, Chapter 5, Section 3.2.1
 - Direction 2399875-800 **LS 3.x** Service Methods -> *Parts Replacement* -> *Console (GOC1 or GOC2)* -> *SRU Replacements - Global Consoles* -> [RIP Board Replacement](#)
 - Direction 5112256-800 **LS4.x** Service Methods -> *Parts Replacement* -> *Console (GOC1 or GOC2)* -> *SRU Replacements - Global Consoles* -> RIP Board Replacement
1. SCSI board should already have four standoffs mounted as illustrated in Figure 2 below.
 2. On the bottom side of the Motorola Board remove and retain the four (4) screws holding the Daughter SCSI board in place.
 3. Remove the Daughter SCSI Card from the Motorola Mother Board and place on the static mat.

PART NUMBER	DESCRIPTION	QUANTITY
2265396	SCSI board	1
46-328421P4	Screws	4

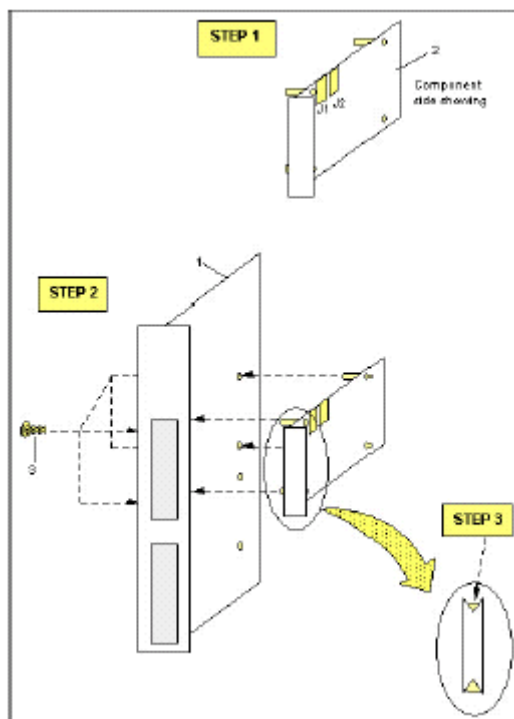


Figure 2 – Removing SCSI Board from the Motorola Board

3.3.2 - UNMOUNTING DIP BOARD FROM MOTOROLA BOARD

NOTE: Do not follow steps in the Illustration (provided here for RIP illustration value). This Illustration shows the proper card placement and screw locations. DO use the following steps:

- Direction 2243314 **LS2.x** System Service Manual, Book 3, Chapter 5, Section 3.2.1
- Direction 2399875-800 **LS 3.x** Service Methods -> *Parts Replacement* -> *Console (GOC1 or GOC2)* -> *HTSRU Replacements - Global Consoles* -> [RIP Board Replacement](#)
- Direction 5112256-800 **LS4.x** Service Methods -> *Parts Replacement* -> *Console (GOC1 or GOC2)* -> *SRU Replacements - Global Consoles* -> RIP Board Replacement

1. DIP board should already have four standoffs mounted as illustrated in Figure 3 below.
2. On the bottom side of the Motorola Board remove and retain the four (4) screws holding the Daughter DIP board in place.
3. Remove the Daughter DIP Card from the Motorola Mother Board and place on the static mat.

PART NUMBER	DESCRIPTION	QUANTITY
2259530	DIP board	1
2209277	Standoff	4
46-328421P4	Screw	4
2280022	Knurled thumbscrews	2

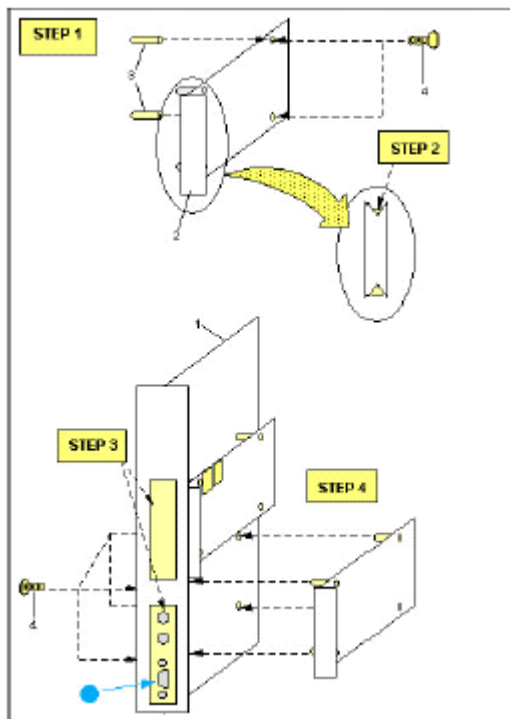


Figure 3 – Unmounting DIP Board From Motorola Board

3.4 - INSTALL DAUGHTER CARDS ONTO NEW MOTOROLA BOARD

3.4.1 MOUNTING SCSI BOARD TO MOTOROLA BOARD

1. Verify SCSI board gasket is removed from front faceplate groove.
Verify jumpers J1 and J2 are installed on SCSI board.
2. SCSI board should already have 4 standoffs mounted as illustrated in Figure 4 below.
3. Remove front panel filler blank(s), if applicable.
4. Mount SCSI board to Motorola board, using the 4 screws {46-328421P4}.

PART NUMBER	DESCRIPTION	QUANTITY
2265396	SCSI board	1
46-328421P4	Screws	4

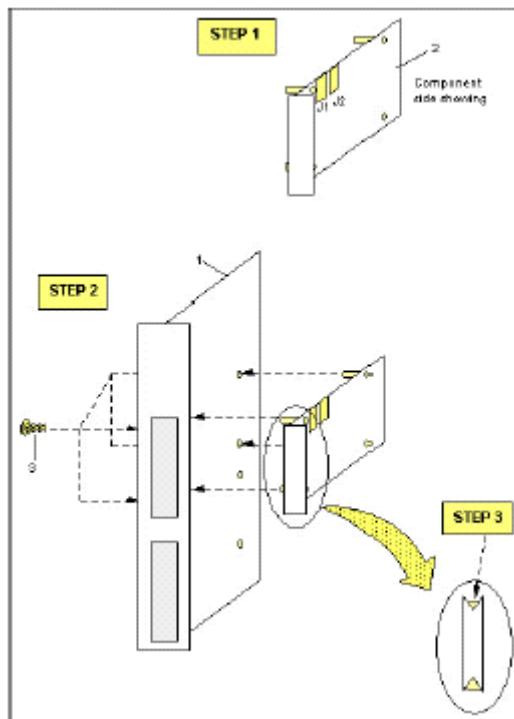


Figure 4 – Mounting SCSI Board to Motorola Board

3.4.2 - MOUNTING DIP BOARD TO MOTOROLA BOARD

1. DIP board should already have 4 standoffs mounted as illustrated in Figure 5 below.
2. Verify DIP board gasket is removed from front faceplate groove.
3. Remove front panel filler blank(s), if applicable.
4. Mount DIP board to Motorola board, using the 4 screws {46-328421P4}.

Verify orientation as shown in the following figure:

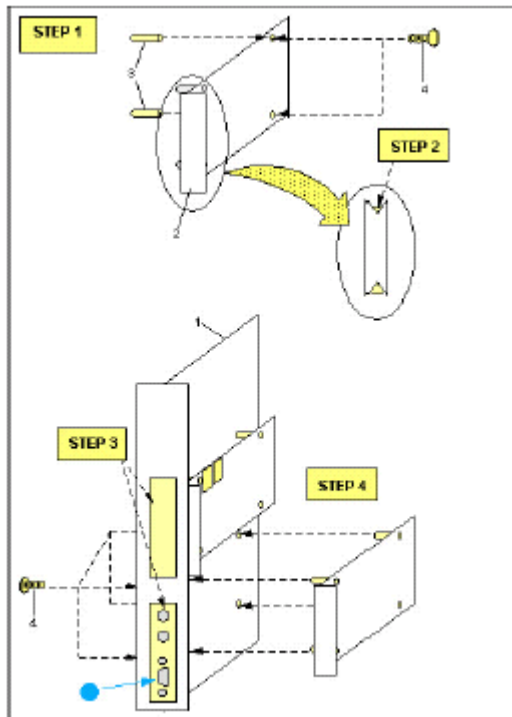


Figure 5 – Mounting DIP Board to Motorola Board

PART NUMBER	DESCRIPTION	QUANTITY
2259530	DIP board	1
2209277	Standoff	4
46-328421P4	Screw	4
2280022	Knurled thumbscrews	2

3.5 - MOUNTING MOTOROLA BOARD ASSEMBLY INTO VME CARD CAGE

Install Motorola board Assembly into VME cage and tighten knurled thumbscrews.

NOTE: Finger tight is appropriate; do not over-torque knurled thumbscrews.

NOTE: When board is properly inserted, all faceplates are flush.

3.6 - CONNECTING CABLES TO MOTOROLA BD ASSEMBLY IN VME CARD CAGE

1. If a plug half is present, verify that the plug is in the DIP Board TX bottom.
2. Connect DEBUG (topmost RJ45 port) cable to Motorola board.
3. Connect 10/100 Base T (center RJ45 port) cable to Motorola board.

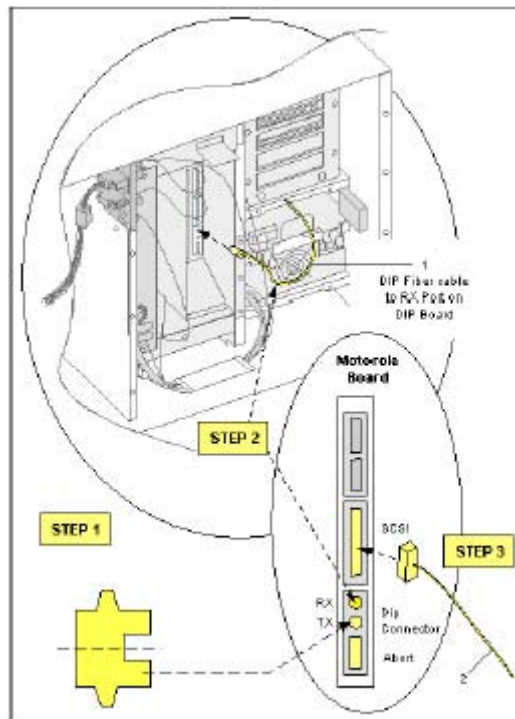


Figure 6 – Connecting Cables To Motorola Board Assembly In The Vme Card Cage

4. Connect SCSI cable A to SCSI Daughter PCI Mezzanine CARD.

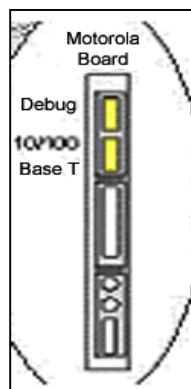


Figure 7 – Connect SCSI cable A to SCSI Daughter PCI Mezzanine CARD

5. Connect DIP fiber cable to top (RX) port on the DIP Daughter board.
6. Connect DIP board DIP Scan Abort cable to DIP Daughter board.
7. Hand-tighten knurled thumbscrews into front panel faceplate of Motorola board (top and bottom).

3.7 - POWER UP (ON) CONSOLE AND PFLASH

1. Power **ON** the console at the front breaker.
2. Stop Applications from auto-starting at the 5 second PINK Box.

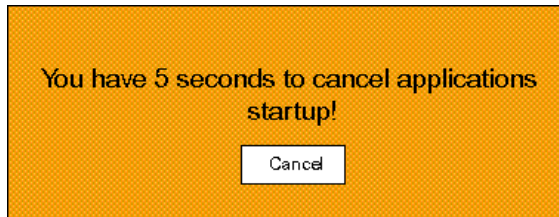


Figure 8 – Power Up (On) The Console And Pflash - 1

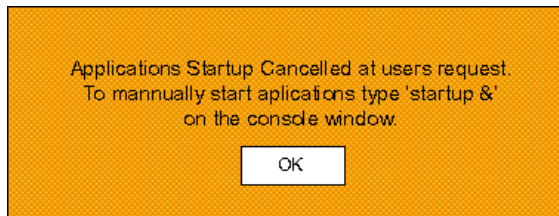


Figure 9 – Power Up (On) The Console And Pflash - 2

3.8 - PFLASH

NOTE: ONLY perform pflash with Applications Down.

1. Open a Unix Shell from the Tool chest.
2. Execute the following command:
Type:
`cu ice`
Press the key.
3. Verify the PPCIBUG prompt is present.
Type:
`~.`
4. Type:
`cd /usr/g/ice/bin`
`pflash`
`/usr/g/ice/bin/sbcutil -info`
See Appendix – Example pFLASH output, for example output.
5. Start up the Application software:
Type:
`st`
6. Close the Unix Shell.
7. If required for your upgrade, proceed with the LFC (see Dir 2348269-100).

APPENDIX – EXAMPLE PFLASH OUTPUT

```
reading board information
Connecting to the ice (serial connection)
We are Connected !!!
debug> timeout=30

debug> setPromptPPCBUG
cnfg
Board (PWA) Serial Number = "5349018      "
Board Identifier           = "MVME2433-4
"
Artwork (PWA) Identifier  = "01-W3839F27A
"
MPU Clock Speed           = "350"
BUS Clock Speed           = "100"
Ethernet Address          = 0001AF0A086D
Primary SCSI Identifier    = "07"
System Serial Number      = "
"
System Identifier         = "
"
License Identifier        = "5349018  "
PPC4-Bug>ver
Debugger/Diagnostics
Type/Revision.....=PPC4/1.2
Debugger/Diagnostics Revision
Date.....=02/15/99  RM01
MicroProcessor
Version/Revision.....=0008/83
02
MicroProcessor Internal Clock Speed
(MHZ).....=350
MicroProcessor External Clock Speed
(MHZ).....=100
Board Type
Identifier.....=MVM
E2400
PCI Bus Clock Speed
(MHZ).....=33
Local Memory
Size.....=10000
000 (256MB)
L2 Cache
(External).....=0
KB
L2 Cache (P0-In-
```

```
Line).....=1024KB
L2 Cache (P1-In-
Line).....=0KB
Super I/O Device
Offset/ID/Revision.....=UNKNOWN
PCI Function 00/00/0 (00000000)
ID/Revision.....=48031057/03
PCI Function 00/0B/0 (00005800)
ID/Revision.....=056510AD/17
PCI Function 00/0B/1 (00005900)
ID/Revision.....=010510AD/17
PCI Function 00/0D/0 (00006800)
ID/Revision.....=000010E3/02
PCI Function 00/0E/0 (00007000)
ID/Revision.....=00191011/41
PCI Function 00/10/0 (00008000)
ID/Revision.....=435417B6/00
PCI Function 00/11/0 (00008800)
ID/Revision.....=00211000/01
PCI Function 00/11/1 (00008900)
ID/Revision.....=00211000/01
PPC4-Bug>
Process Complete !
/usr/g/ice/bin/sbcutil -env
setting PPCBug NVRAM parameters

Connecting to the ice (serial connection)
We are Connected !!!
debug> timeout=30
debug> setPromptPPCBUG
debug> timeout=15

Process Complete !
/usr/g/ice/bin/sbcutil -bs
FLASHing bootstrap to allow dual FLASH boot

Connecting to the ice (serial connection)
We are Connected !!!
debug> timeout=30
debug> setPromptPPCBUG
debug> timeout=15

Process Complete !
/usr/g/ice/bin/sbcutil -br
FLASHing VxWorks bootrom
```

```
Connecting to the ice (serial connection)
```

```
We are Connected !!!
```

```
debug> timeout=30
```

```
debug> setPromptPPCBUG
```

```
inetEthernet=192.9.220.11
```

```
debug> timeout=90
```

```
tftp vxWorks BootRom Code into Memory...
```

```
debug> timeout=15
```

```
debug> timeout=15
```

```
Performing Board Reset
```

```
debug> timeout=15
```

```
Process Complete !
```

```
/usr/g/ice/bin/sbcutil -config
```

```
setting VxWorks NVRAM parameters
```

```
Connecting to the ice (serial connection)
```

```
We are Connected !!!
```

```
debug> timeout=30
```

```
debug> setPromptVxWorksBoot
```

```
debug> bootConfigVxWorks
```

```
Process Complete !
```

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